

LLM & GENAI

- vLLM
- Unsloth
- LoRA / QLoRA
- PEFT
- Transformers
- LLMOps / Self-hosted Serving
- LLM Evaluation

MACHINE LEARNING

- XGBoost
- Autoencoders
- PyTorch
- TensorFlow
- Keras
- Scikit-learn
- DenseNet
- Logistic Regression
- Classification
- Predictive Modelling
- Unsupervised Learning
- Feature Engineering
- Transfer Learning
- Anomaly Detection
- NLP
- Computer Vision

GPU STACK

- CUDA GPU Libraries
- RAPIDS cuDF / CuPy
- Multi-GPU Training

OPTIMISATION & FORECASTING

- Google OR-Tools
- Vehicle Routing
- Mixed-Integer Programming
- Operations Research
- Constraint Modelling
- Assignment Models
- Supply-Chain Optimization
- Scenario Analysis
- Time-Series Forecasting

PRODUCTION ML PLATFORM

Cloud & Managed ML

- GCP
- GKE
- Vertex AI
- Vertex AI Feature Store
- GCS
- Artifact Registry
- Azure ML
- Function Apps
- Logic Apps

Serving & Orchestration

AI/ML engineer with 10+ years taking models from research notebooks to reliable customer-facing systems. Designs, implements, and maintains end-to-end ML systems across data preparation, training, evaluation, deployment, monitoring, controlled releases, and rollback. At Best Buy, built a config-driven pipeline that fine-tunes open-source language models with LoRA and serves them in-house, replacing paid external AI APIs with estimated annual API-cost savings of **~\$300K** while supporting fraud systems with feature pipelines handling 40,000 events/sec at peak. At Wipro, built GPU-accelerated cybersecurity ML on an NVIDIA DGX supercomputer (8×A100). At Accenture, developed an Azure ML product-quality system that saved \$2.5M/year and designed Google OR-Tools systems for vehicle routing, transport planning, fleet capacity, delivery time windows, waiting time, load distribution, and store demand. Also built supply-chain simulations comparing logistics cost, product freshness, carbon impact, and hub-location scenarios, and earlier delivered time-series forecasting across 15+ industry verticals. Lead the design and delivery of AI/ML platforms, mentors engineers, and supports technical hiring across ML engineering teams.

WORK EXPERIENCE

- Lead Artificial Intelligence Engineer

Best Buy · Bengaluru, India

Jul 2025 – Present

  - Built a config-driven **LoRA fine-tuning framework** with Unsloth, PEFT, and Transformers; enabled YAML-driven task addition and single- or multi-GPU training.
  - Design, implement, and maintain end-to-end ML systems spanning data pipelines, training, evaluation, registry, serving, monitoring, controlled releases, and rollback.
  - Deployed **vLLM** serving on GKE with HPA autoscaling, Istio routing, GCS model storage, Artifact Registry, Cloud Build, and Prometheus/Grafana observability.
  - Built fraud systems across batch, streaming, and real-time models. Beam/Dataflow feature pipelines handled **40,000 events/sec** at peak; supported a **200 ms scoring SLA** with secure, monitored GKE/Seldon Core inference and load testing.
- Lead ML Engineer — ML Engineering & Operations

Wipro · Client: Major US Retail Corporation

Aug 2023 – Jul 2025

  - Built behavioural anomaly detection for AWS account security using per-entity autoencoders on **NVIDIA DGX (8×A100)** with NVIDIA Morpheus and billion-row CloudTrail datasets.
  - Developed phishing detection processing **2,400 emails/sec** and RAPIDS data pipelines for GPU-accelerated training and inference; monitored model and system health through MLflow and Grafana.
  - Led ML/MLOps delivery and mentoring at Wipro; completed **300+ résumé screens** and **50+ technical interviews**.
- Senior Data Scientist — Data Science / Azure ML

Accenture

May 2021 – Aug 2023

  - Developed an end-to-end Azure ML product-quality pipeline for BamaGruppen, Norway—**\$2.5M annual savings**, inspections increased from **130 to 160 trucks/day**, and defect detection improved **17%**.
  - Designed Google OR-Tools vehicle-routing models for daily pallets-to-stores delivery under vehicle-count, capacity, time-window, waiting-time, and store-demand constraints; integrated Azure Functions, Logic Apps, and RPA for automated dispatch.
  - Built a Google OR-Tools MIP system for daily transport departures and supply-chain simulations comparing operating cost, product freshness, carbon impact, and regional hub-location scenarios; owned deployment, monitoring, and CI/CD.

KubernetesSeldon Core

DockerHelmIstio

Kubernetes HPA

DATA & ML LIFECYCLE

Apache BeamDataflow

Pub/SubMLflow

Streaming Pipelines

Training Pipelines

Automated Evaluation

Experiment Tracking

Model Registry

Model Versioning

Continuous Training

RELIABILITY, DELIVERY & LEADERSHIP

CI/CDGitOpsCloud Build

ArgoCDGitHub Actions

PrometheusGrafana

Blue-Green Delivery

Model MonitoringRollback

Load TestingAlerting

Architecture ReviewsMentoring

Stakeholder Alignment

LANGUAGES, DATA & DATABASES

PythonSQLPandas

NumPyMS SQL Server

IBM DB2KQL

CERTIFICATIONS +

Oracle Cloud Infrastructure 2024  
Generative AI Certified Professional

Microsoft Certified  
Azure AI Engineer Associate

Microsoft Certified  
Azure Data Scientist Associate

Microsoft Certified  
Power BI Data Analyst Associate

EDUCATION

B.E. Electrical & Electronics

Andhra University, Visakhapatnam  
2012 – 2016 · 80%

ACHIEVEMENTS

Best Buy Talent Quotient (2026):  
cybersecurity and fraud delivery.

- Data Scientist

GD Research Centre Pvt Ltd

May 2020 – May 2021

Designed and deployed domain-aware time-series forecasting across **15+ industry verticals**, with feature-engineering pipelines for heterogeneous data and long-horizon business forecasts.
- Deep Learning Engineer

Tata Consultancy Services

Jul 2019 – Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours**, with **90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.
- Senior Software Engineer — Machine Learning

Infosys

May 2016 – Jul 2019

Built an ML-based lead-conversion prediction system and business-intelligence applications, following earlier production software work on mainframe systems.

PROJECTS

LLM Fine-tuning & Serving Framework

Best Buy · Jul 2025 – Present

Config-driven LoRA adapter training for open-source causal LLMs with Unsloth and PEFT/Transformers, automated evaluation, GCS artifact lifecycle, and vLLM serving on GKE with autoscaling, Istio routing, and production monitoring.

~\$300K estimated annual API-cost savings saved

Vehicle Routing & Transport Planning

Accenture · May 2021 – Aug 2023

Designed Google OR-Tools VRP and MIP models for daily delivery and terminal-to-distribution-centre planning. Modelled vehicle, capacity, time-window, waiting-time, demand, and departure constraints, then automated operational use through Azure Functions, Logic Apps, RPA, and Slack notifications.

VRPMIPAzure Functions

Automated daily transport plans · Replaced manual spreadsheet planning

Supply-Chain Optimization

Accenture · May 2021 – Aug 2023

Built a simulation and optimisation framework covering vendors, warehouses, and customers. Compared operating cost, product freshness, carbon footprint, and network scenarios to identify the appropriate number and locations of regional distribution hubs.

OR-ToolsSimulationPower BI

Informed regional hub-location and distribution decisions

Industry Forecasting Platform

GD Research Centre · May 2020 – May 2021

Designed and deployed domain-aware time-series forecasting for GlobalData UK across energy, manufacturing, FMCG, healthcare, finance, oil and gas, retail, technology, pharmaceuticals, mining, aerospace, telecommunications, automotive, travel, and other sectors.

Time Series

15+ industry verticals · Enterprise market-intelligence forecasts

Product Quality Prediction

Accenture · BamaGruppen, Norway

Led an end-to-end Azure ML system that predicted product quality at warehouse arrival, automated inspection prioritisation, and integrated model outputs with operational workflows and monitoring.

Azure Pipelines

\$2.5M/year saved · 130 to 160 inspections/day · 17% higher detection

Real-Time Fraud Detection Platform

Best Buy · Jul 2025 – Present

Designed production pipelines from streaming feature generation through model serving, observability, controlled deployment, and rollback for retail fraud decisions across payments, refunds, reshipping, and chargebacks.

40k feature events/sec · 200 ms scoring SLA

ADDITIONAL PROJECTS

- AWS anomaly detection:

per-entity autoencoders on billion-row CloudTrail data with Morpheus, RAPIDS, MLflow, and an 8×A100 DGX.
- Semiconductor defect classification:

DenseNet on about 1 million wafer images, three-hour training cycles, and South Korea deployment.